

TWIN-64

Architecture Document

Helmut Fieres
June 27, 2025

Contents

Introduction	1
Concepts	1
This Book	1
Architecture Overview	3
System Organization	5
Data Types	5
General Registers	5
Control Registers	5
Processor Status Register	5
Byte Ordering	5
TLB	5
Caches	5
System Memory	7
Virtual Memory	7
Physical Memory	7
Addressing and Access Control	9
Address Translation	9
Access Control	9
Access Protection	9
Interrupts	11
Traps	11
External Interrupts	11
Instruction Set	13
Instruction Overview	13
Instruction Formats	13
ADD - Integer Addition	16

Introduction

Concepts

This book

Architecture Overview

System Organization

Data Types

General Registers

Control Registers

Processor Status Register

Byte Ordering

TLB

Caches

System Memory

Virtual Memory

Physical Memory

Addressing and Access Control

Address Translation

Access Control

Access Protection

Interruptions

Traps

External Interrupts

Instruction Set

Instruction Overview

Integer Arithmetic Instructions

ADD, SUB, SHLxA, SHRxA

Bit Operations Instructions

AND, OR, XOR, EXTR, DEP, DSR

Immediate Instructions

LDI, ADDIL, LDO

Memory Reference Instructions

LD, ST, LDR, STC

Branch Instructions

B, BR, BV, BB, CBR, MBR

Combined Instructions

ADD, SUB, AND, OR, XOR, CMP

System Instructions

MFCCR, MTCR, RSM, SSM, RFI, ITLB, PTLB, PCA, FCA, LDPA, PRBR, PRBW

Instruction Formats

T64 uses few instruction formats. They are grouped by the number of available register slots and the length of the immediate value field.

Immediate S19

The immediate S19 instruction format provides one register and a 19 bit signed immediate field. This format is typically used by branch instructions.

Grp	OpCode	Reg R	Opt 1	S-IMM-19				
2	4	4	3	19				

Immediate S15

The immediate S15 instruction format provides two registers and a 15 bit signed immediate field. This format is used by branch instructions as well as instruction which operate on a register and an immediate value.

Grp	OpCode	Reg R	Opt 1	Reg B	S-IMM-15			
2	4	4	3	4	15			

Immediate S13

The immediate S13 instruction format provides two registers, an additional option field and a 13 bit signed immediate field. Some instruction use the S-IMM-13 field for special encodings instead of a signed value. This format is used by memory reference instructions where the address is formed by a base register and a 13-bit signed offset.

Grp	OpCode	Reg R	Opt 1	Reg B	Opt 2	S-IMM-13 / special		
2	4	4	3	4	2	13		

Immediate S9

The immediate S9 instruction format provides three registers, an additional option field and a 9 bit signed immediate field. Some instruction use the S-IMM-9 field for special encodings instead of a signed value. This instruction format is used for all computational instructions where three registers are needed.

Grp	OpCode	Reg R	Opt 1	Reg B	Opt 2	Reg A	S-IMM-9 / special	
2	4	4	3	4	2	4	9	

Immediate U20

The immediate U20 format is used by the immediate instruction group to provide a 20bit unsigned value. This value is then placed at certain positions in the register target Reg R.

Grp	OpCode	Reg R	0	U-IMM-20
2	4	4	2	20

ADD Integer Addition

Syntax

```
ADD RegR, RegB, RegA
ADD RegR, RegB, Immed15
ADD RegR, Immed13( RegB )
ADD RegR, RegX ( RegB )
```

Format

The ADD instruction uses the register, the immediate, the indexed and the register indexed instruction formats.

0	1	Reg R	0	Reg B	Opt 2	Reg A	S-IMM-9 / special
2	4	4	3	4	2	4	9

0	1	Reg R	0	Reg B	S-IMM-15
2	4	4	3	4	15

1	1	Reg R	0	Reg B	dw	S-IMM-13
2	4	4	3	4	2	13

1	1	Reg R	0	Reg B	dw	Reg X	0
2	4	4	3	4	2	4	9

Description

The ADD instruction adds two signed 64-bit operands and stores them to the target register **RegR**.

The immediate instruction format will add the sign extended 15-bit value to **RegB**. The register instruction format instruction will add **RegA** and **RegB**.

The indexed instruction format will load the content of memory address formed by adding the scaled immediate 13-bit field to the base register **RegB** and add the data value to **RegA**. The **dw** field indicates the data value width. A **dw** value of zero load an 8-bit, a value of 1 a 16-bit value, value of 2 a 32-bit value and a value of 3 a 64-bit value. In any case, the data value fetched is signed extended to a 64-bit value. In addition, the **dw** field will scale the offset value by left shifting the immediate field according to the data width.

The register indexed instruction format will load the content of memory address formed by adding **RegX** to the base register **RegB** and add the data value to **RegA**. Both load formats will use the **dw** field to specify the data value width. A **dw** value of zero load an 8-bit, a value of 1 a 16-bit value, value of 2 a 32-bit value and a value of 3 a 64-bit value. In any case, the data value fetched is signed extended to a 64-bit value. The offset in **RegX** is interpreted as a byte offset, independent of the actual **dw** field value.

Operation

$\text{RegR} \leftarrow \text{RegB} + \text{RegA}$ (register format)
 $\text{RegR} \leftarrow \text{RegB} + \text{immOperand}()$ (immediate format)
 $\text{RegR} \leftarrow \text{RegR} + \text{memOperand}()$ (indexed formats)

Exceptions

Integer Overflow
Data Alignment
Memory Access Violation
Memory Protection Violation
Data TLB miss

Notes

None.

